

the Semisimplicity conjecture. Moreover, all rational maps on Y satisfies the Dynamical degree comparison conjecture.

An explicit example is the Hilbert scheme $X^{[2]}$ of zero cycles of length 2 of appropriate varieties X (see Section 2 for more detail).

For proving Theorem 1.1, we combine the ideas in [HT24] to that in [Xie24]. The first main idea in [Xie24] is, over a finite field $\mathbb{F}_q$, to each finite correspondence f and each $k = 0, 1, \dots, n$, consider a composition $Fr_q^s \circ f^t$ where s is an integer (it is important to allow **negative values for** s) and t a positive integer. If the dynamical degrees $\lambda_k(f)$ are a log-concave sequence, using that Frobenius morphisms commute with correspondences defined over the same finite field, with a suitable choice of s and t one can make sure that $\lambda_k(Fr_q^s \circ f^t)$ is not too far away compared to $\max_{j=0, \dots, n} \lambda_j(Fr_q^s \circ f^t)$. Note that the choice of μ_i in the proof of Lemma 2.13 in [Xie24] is the same as the choice of r in the proof of Lemma 4.9 in [HT24]. Note that for a finite correspondence f , as in the case of endomorphisms, the cohomological dynamical degree $\chi_{2k}(f)$ is the same as the spectral radius of $f^*|_{H^{2k}(X)}$. Therefore, a known result on spectral radii comparison (Theorem 1.2 in [Tru24]), which is a generalisation of Entropy comparison conjecture, can be used to show that $\chi_{2k}(f) = \lambda_k(f)$. The inequality $\chi_{2k+1}(f)^2 \leq \lambda_k(f)\lambda_{k+1}(f)$ is obtained from this by the trick of using the product correspondence $f \times f : X \times X \rightarrow X \times X$. Thus, Dynamical degree comparison conjecture holds for finite correspondences whose numerical dynamical degrees are a log-concave sequence. In the general case, the numerical dynamical degrees $\lambda_k(f)$ may not be log-concave. The second main idea in [Xie24] is (see Theorem 1.14 therein, and Proposition 1.4 below) to establish some interesting relations between certain polytopes of two sequences satisfying certain properties. Theorem 1.14 in [Xie24] can be viewed as a continuous version of Lemma 4.9 in [HT24] (see Proposition 1.4 below).

To prove Theorem 1.1, here we replace the cohomological operator G_r in the approach in [HT24] by composing with iterations of F^* . There is a complication arising in case f is not a finite correspondence: in this case the dynamical degree is not the spectral radius but involves norms of dynamical iterates of the correspondence $f^{\diamond t}$. In case the dynamical degrees $\lambda_k(f)$ are a log-concave sequence, then again Lemma 4.9 in [HT24] can be used to conclude Theorem 1.1. In the general case, we make use of special properties of the Frobenius norm on matrices and of asymptotic singular values of powers of a matrix (this property was used in [Hu19, Hu24] to solve Dynamical degree comparison conjecture for endomorphisms of Abelian varieties), as well as use essentially the second main idea in [Xie24] mentioned above, but in the following form which is similar to the form of Lemma 4.9 in [HT24]. (More precisely, in Lemma 4.9 in [HT24], Condition 2 below is required, but instead of Condition 1 below it is required that the sequence a_j is log-concave.)

Proposition 1.4. *Let $a_0, a_1, \dots, a_n$, and $b_0, b_1, \dots, b_{2n}$ be non-negative numbers. Assume the following two conditions are satisfied:*

Condition 1: $b_{2j} \geq a_j$ for all $j = 0, \dots, n$.